Snapshot Ecosystem & Socioeconomic Profile (ESP)

A working paper for the 2026 Longfin Squid Research Track Assessment (TOR1)

Stephanie Owen, NEFSC EDAB (Affiliate)
August 3, 2026

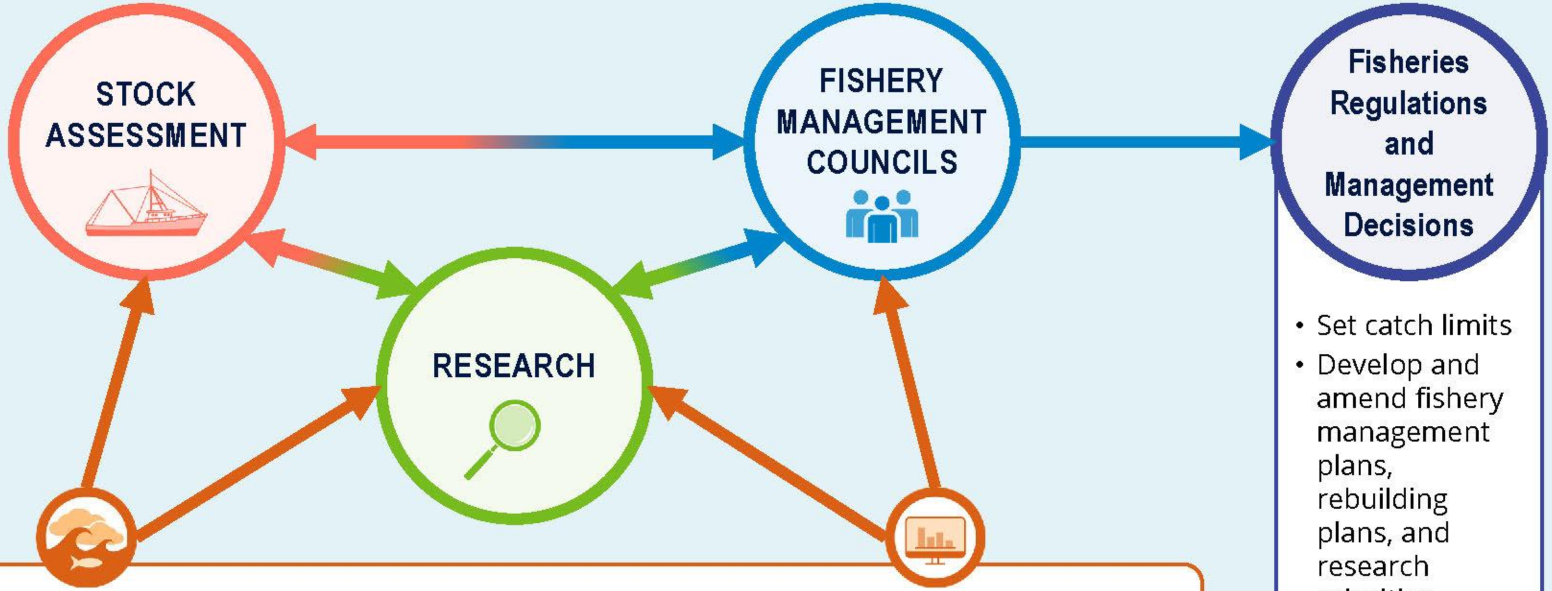
ESP Objectives

- Leverage existing information and knowledge pathways
  - Incorporate a broad range of information
  - Identify cumulative and comprehensive patterns
- Facilitate interpretation and use in management
  - Standardized framework & visuals
  - Improve transparency, reproducibility, and efficiency
- Identify on-ramps to fill knowledge gaps and work toward operational Ecosystem Based Fisheries Management
  - Provide relevant ecosystem and socioeconomic information for fisheries management
- Track changes in the system over time

Image courtesy of Rebecca White, AFSC Communications Branch

NOAA FISHERIES

FISHERIES SCIENCE AND MANAGEMENT SYSTEM



- Set catch limits
- Develop and amend fishery management plans, rebuilding plans, and research priorities

Ecosystem and Socioeconomic Profile Information

Addresses Term of Reference to identify ecosystem and climate influences

- Improves fisheries stock assessments
- Informs decisions that determine the health and abundance of fish stocks

Provides scientific information to fisheries managers

- Improves estimates of risk and uncertainty
- Can be used in setting sustainable harvest policies
- Highlights research needs and data gaps

Integrating ESPs into the existing stock assessment and management schedule

- **Full ESPs** have historically been presented with Research Track Stock Assessments
  - Opportunity to test ecosystem covariates in the stock assessment model
- **Snapshot ESPs** can be presented with a stock assessment or following operational model peer review
  - Communicate recent indicator status and recent events/ observations that could be contextually important for the stock
  - Intended to be easily updatable for management tracks or data updates

Longfin Squid Snapshot ESP

- Longfin squid life history and conceptual model
- Highlights from the 2025 Risk Assessment
  - Risks to the stock and commercial fishery
- Ecosystem and Socioeconomic indicators
  - Implications on the fishery and habitat suitability for most recent year
  - Areas of uncertainty to drive future research
- Reproducible in the future with few changes to incorporate new data updates

Longfin Inshore Squid (*Doryteuthis pealeii*) Snapshot Ecosystem & Socioeconomic Profile (ESP)

August 2026

Key Findings from the Life History Working Group

Lifespan and aging

Some literature sources estimate growth to be 1 statolith ring/day, and literature review supports a lifespan of less than 1 year. Participants at the Longfin Inshore Squid Summit estimated a maximum age of 15 months. Statolith aging from the Squid Biological Sampling Program (SQUIBS; 2023-2025) indicates a maximum age of 264 days from squid caught in the fishery.

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Migration and movement dynamics

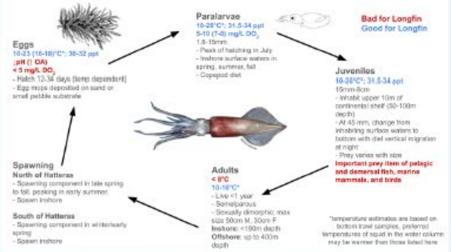
In November/December, longfin squid migrate from the inshore shelf to deep, warm slope waters to overwinter. By May/June of the next year, they migrate back to shallow coastal shelf waters to spawn [1]. The fishery follows this pattern, occurring offshore in winter and inshore in summer. Recent work hypothesizes a winter cohort that hatches south of Cape Hatteras and migrates onto the Northeast shelf [2]. Fishery observations describe a spatial gradient of 1-6 cm mantle length (ML) squid from waters south of Hatteras through southern New England, with the smallest squid detected further south. The Gulf Stream and warm core rings may facilitate the recruitment transport of juvenile squid, but potential for inputs to the population from the South and offshore are difficult to quantify.

Reproductive dynamics

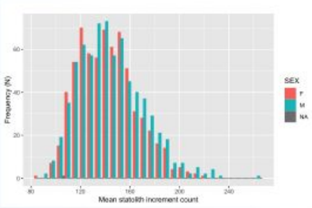
Spawning peaks inshore from late spring to early summer in the Mid-Atlantic and southern New England [3] [1] with hatching in late summer [4]. Consideration of the hypothesis of a winter cohort spawning south of Hatteras indicates the presence of multiple cohorts with some outside of the traditional Northeast shelf stock area, and provides evidence of year-round spawning.

Natural mortality

Although natural mortality is expected to be age-dependent, lack of accurate age data makes further study difficult. Using the equation derived by Hamel and Cope [5], natural mortality for longfin squid can range from 0.36 (max. age = 15 mo.) to 0.675 (max. age = 8 mo.). Cannibalism impacts natural mortality, but there is no available data to quantify the amount of mortality this causes.



Age Frequency from SQUIBS

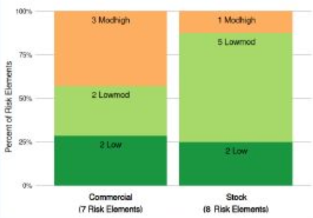


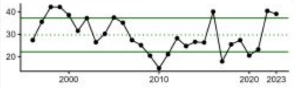
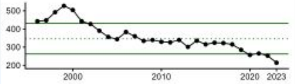
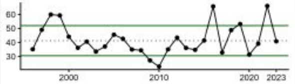
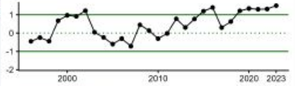
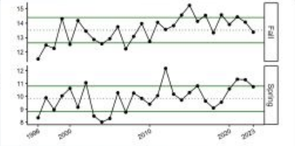
Key Points from the Mid-Atlantic Risk Assessment

The Mid-Atlantic Fishery Management Council (MAFMC) Ecosystem Approach to Fisheries Management (EAFM) Update [6] determined that there are moderate-high risks related to:

- Potential and observed distribution shifts of longfin squid
- Not achieving optimum yield due to interactions with non-MAFMC managed species
- Regulatory complexity: occasional recent changes in regulations and moderate (3-4) recreational regulation differences across states. Compliance is challenging for fishery participants given complex regulations.
- Discards: While various management measures have minimized discards to the extent practicable, ecosystem changes may affect existing measures.

Risk elements are aspects that may threaten achieving the biological, economic, or social objectives to achieving optimal yield that the MAFMC desires from a fishery. Longfin squid did not score in the "high" risk category for any risk elements in 2025.



Indicator Units	Status In 2023	Implications	Time Series
Commercial landings (millions of lbs)	Above long term (1996-2023) average	Environmental dynamics vary between locations/timing of the summer and winter squid fisheries. An increase in landings since 2020 but decrease in number of vessels could indicate targeted trips in specific times of year and fishers targeting other species when longfin are not available.	
Number of commercial vessels (# of federally-permitted vessels landing over 1lb of longfin squid)	Below long term (1996-2023) average	Number of commercial vessels has been steadily decreasing since around 2000 consistent with decreasing fleet diversity and continued risk to fishery resilience [7]. Permit requalification in 2019 and a decrease in the post-closure trip limit for trimester 2 may cap participation, although these actions were designed to accommodate recent fishing trends and activity.	
Commercial revenue (millions, inflation adjusted to 2025 USD)	Near long term (1996-2023) average	Commercial revenue has decreased since 2022, driven by the overall decrease in landings by 23% [7].	
Western Gulf Stream Index (shift in the western part of the Gulf Stream North wall: mean position: >0 = more northerly, <0 = more southerly)	Above long term (1996-2023) average	Since the mid-1990s, north and westward shifts in the Gulf Stream have resulted in an increase in warm core rings and deep water, high salinity heat waves. The position of the Gulf Stream influences seasonal temperature and water mass mixing dynamics that affect longfin squid habitat suitability, temperature-dependent growth, and prey availability (https://noaa-edab.github.io/catalog/gsi.html).	
Bottom temperature in Mid-Atlantic Bight (MAB) and Southern New England (SNE)(°C)	Near long term (1996-2023) average (Fall and Spring)	Inshore temperature thresholds (around 14°C) initiate migration of squid from offshore overwintering habitats. Longfin squid seasonal distribution and growth rates are likely temperature dependent, avoiding water <8°C. Stronger and/or more persistent Mid-Atlantic Cold Pool conditions (not shown) may limit habitat availability (https://noaa-edab.github.io/catalog/cold_pool.html).	

* [7] — Longfin Squid Fishery Information Document

Data Gaps/Uncertainty

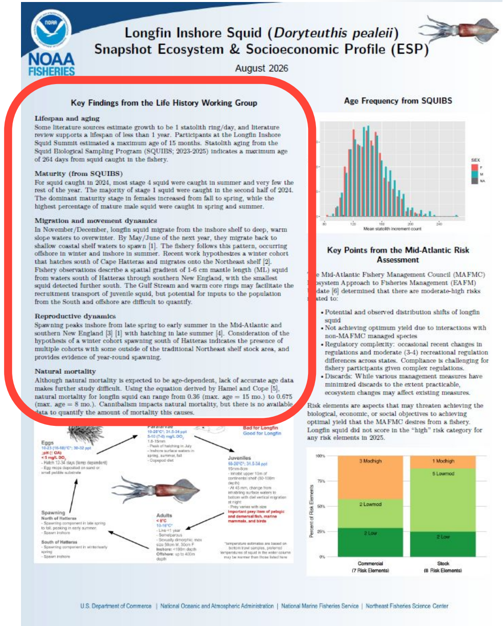
- Bottom temperature data comes from GLORYS [8], a modeled re-analysis product that incorporates insitu data.
- The Western Gulf Stream Index indicator is a yearly value and may not be indicative of changes in oceanographic processes on a smaller time scale.
- Aging uncertainty creates uncertainty around life history processes, spawning location and timing, and natural mortality.
- While survey data in the 1970s and 80s indicate larval squid south of Cape Hatteras in the winter months that are transported north into the Mid-Atlantic Bight, there is a lack of definitive data to assess this hypothesis.
- Effects of cannibalism on the population are unknown at this time.

We welcome your observations! Please contact northeast.ecosystem.highlights@noaa.gov with any on-the-water insights or changes observed in the longfin inshore squid fishery and nefsc.esp.leads@noaa.gov with questions or comments on the information presented in this report.

The code used to create this report can be viewed online: github.com/NEFSC/READ-EDAB-longfinESP

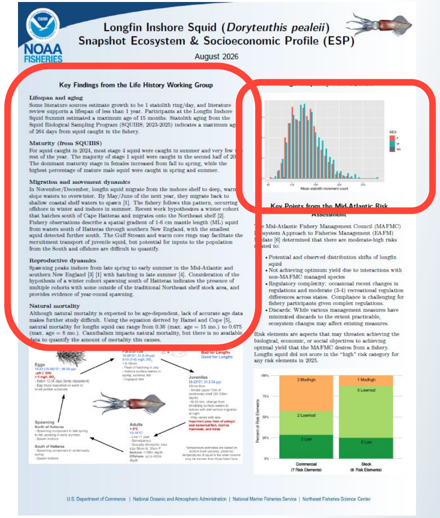
Longfin Squid Life History

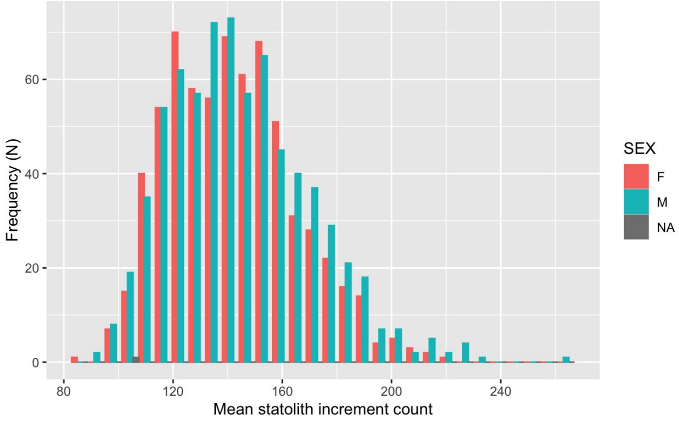
- Working group subgroup and literature review (See also: Owen 2026a - WP1)
- 5 key topics:
  - Lifespan and aging
  - Maturity
  - Migration and movement
  - Reproduction
  - Natural mortality



Life History - Lifespan/Aging

- Lifespan likely less than 1 year supporting hypothesis of 1 statolith ring per day
- Squid Biological Sampling Program (SQUIBS - Mercer et al. 2026b - WP3) squid caught from the fishery had a max statolith count of 264 days
    - Highest frequency of squid with mean statolith counts ~140 for both males and females





Life History - Maturity

- SQUIBS sampling (See Mercer et al. 2026b - WP3)
- For squid caught in 2024:
  - Most stage 4 squid (male) were caught in summer
  - The majority of stage 1 squid were caught in the second half of 2024.
  - The dominant maturity stage in females increased from fall to spring
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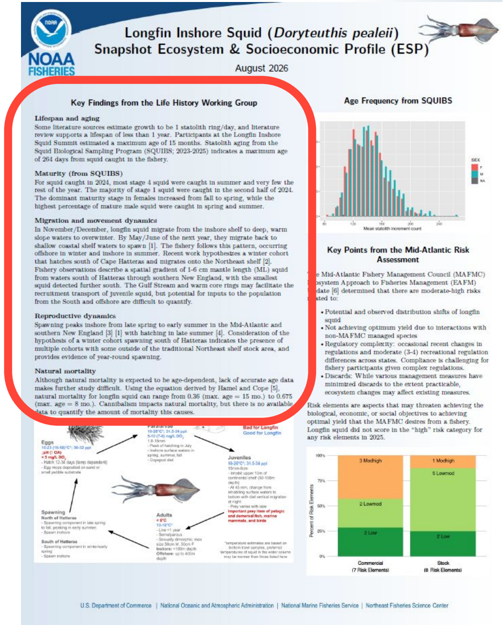

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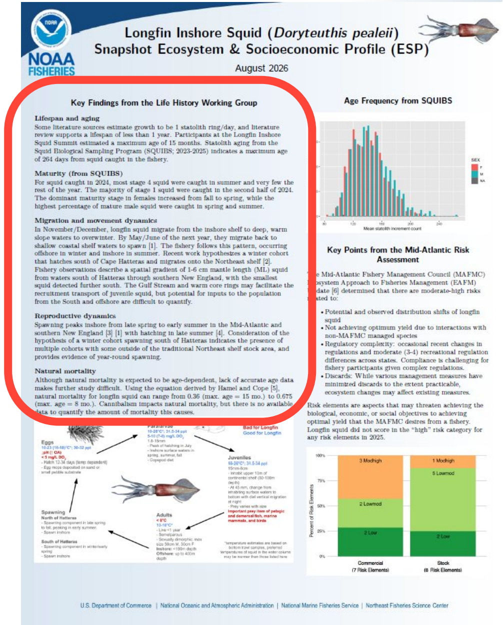
Life History - Migration

- Migrate to warm slope waters to overwinter offshore; move back to coastal shelf waters in May/June to spawn
- Richardson 2025 (WP 4) hypothesizes a winter cohort that hatches south of Hatteras and migrates onto the Northeast Shelf
    - Northward transport may be facilitated by oceanographic processes, but inputs to the NW Atlantic population from the South Atlantic are difficult to quantify



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Mean statolith increment count

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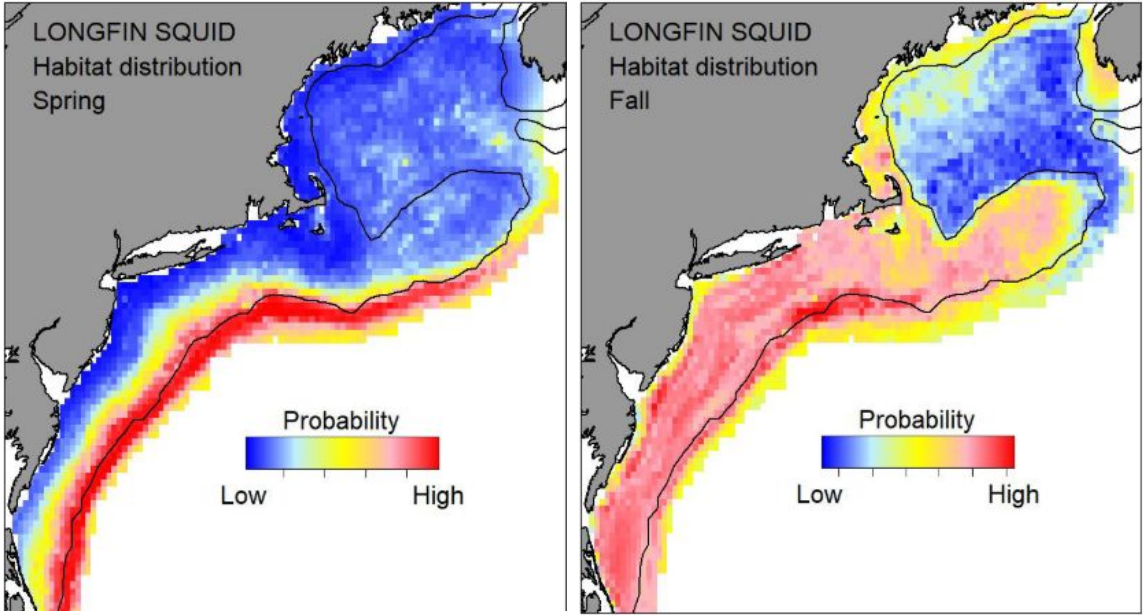


Image credit: NEFSC Ecosystem Dynamics and Assessment Branch

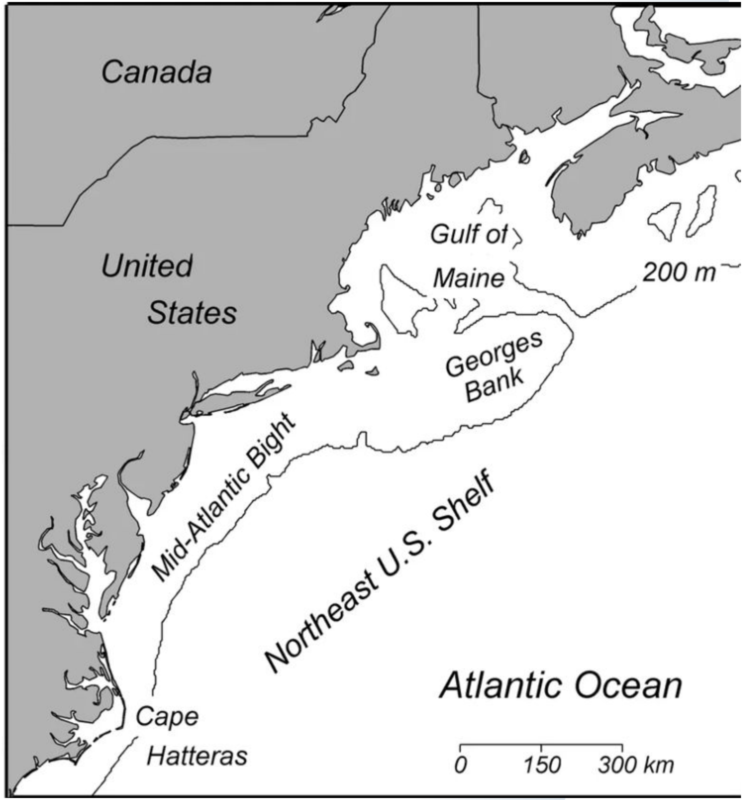
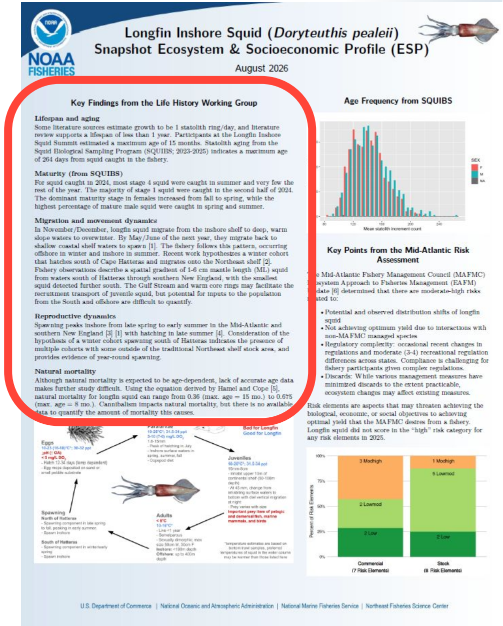


Image credit: Hare et al. (2014)

Life History - Reproduction

- Spawning peaks inshore late spring to early summer, with most hatching in late summer.

- Winter cohort hypothesis (WP4) indicates presence of multiple cohorts outside the NES and year-round spawning



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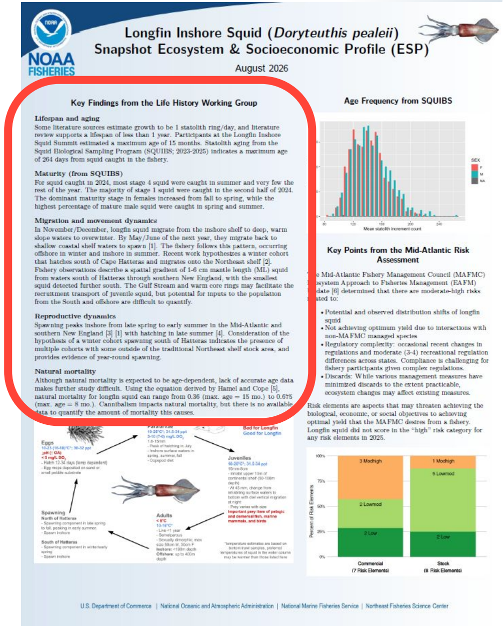
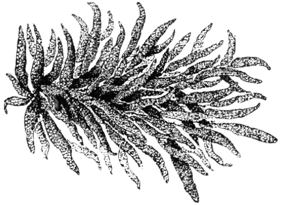
Life History - Natural Mortality

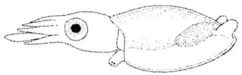
- Lack of accurate age data impedes natural mortality considerations
- Using equations in Hamel and Cope (2022), longfin squid natural mortality could range from 0.36 (max age of 15 months - from Longfin Summit) to 0.675 (max age of 8 months - SQUIBS)
- Cannibalism evident but hard to quantify





Eggs
10-23 (16-18)°C*; 30-32 ppt
↓pH (↑ OA)
< 5 mg/L DO_2
- Hatch 12-34 days (temp dependent)
- Egg mops deposited on sand or small pebble substrate

Paralarvae

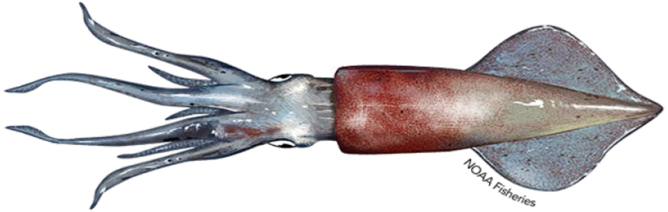


10-26°C*; 31.5-34 ppt
5-10 (7-8) mg/L DO_2
1.8-15mm
- Peak of hatching in July
- Inshore surface waters in spring, summer, fall
- Copepod diet

Bad for Longfin
Good for Longfin

Juveniles
10-26°C*; 31.5-34 ppt
15mm-8cm
- Inhabit upper 10m of continental shelf (50-100m depth)
- At 45 mm, change from inhabiting surface waters to bottom with diel vertical migration at night
- Prey varies with size

Important prey item of pelagic and demersal fish, marine mammals, and birds



Adults
< 8°C
10-16°C*
- Live <1 year
- Semelparous
- Sexually dimorphic; max size 50cm M, 30cm F

Inshore: <180m depth
Offshore: up to 400m depth

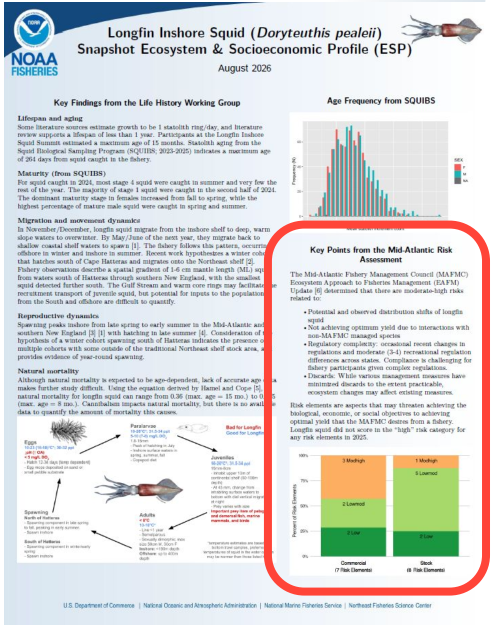
Spawning
North of Hatteras
- Spawning component in late spring to fall, peaking in early summer.
- Spawn inshore

South of Hatteras
- Spawning component in winter/early spring
- Spawn inshore

*temperature estimates are based on bottom trawl samples, preferred temperatures of squid in the water column may be warmer than those listed here

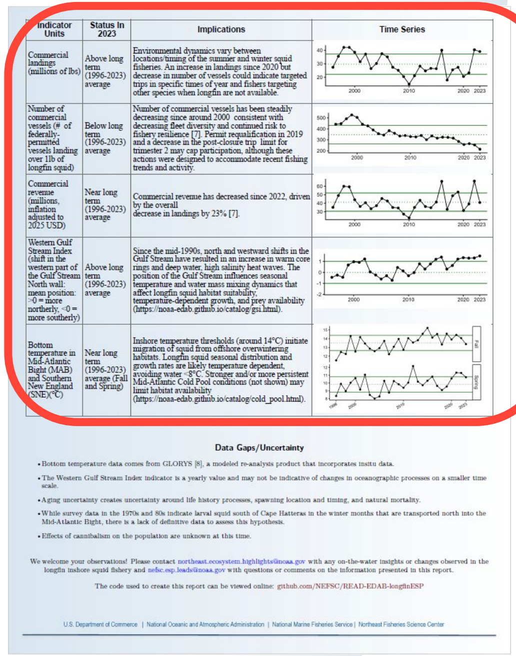
2025 MAFMC Risk Assessment

- "Risk elements are aspects that may threaten achieving biological, economic, or social objectives that prevent optimal yield."
- Moderate-high risks:
  - Potential and observed distribution shifts
  - Interactions with non-MAFMC managed species
  - Occasional regulatory changes and moderate recreational regulation differences between states
  - Discards, although management measures in place to minimize
- No high risk elements in 2025!



Ecosystem and Socioeconomic Indicators

- Indicator time series - ending in 2023 (last year of data in the assessment)
- Status relative to long-term mean
- Implications to the stock and/or fishery



Indicator Units	Status In 2023	Implications	Time Series
Commercial landings (millions of lbs)	Above long term (1996-2023) average	Environmental dynamics vary between locations/timing of the summer and winter squid fisheries. An increase in landings since 2020 but decrease in number of vessels could indicate targeted trips in specific times of year and fishers targeting other species when longfin are not available.	
Number of commercial vessels (# of federally-permitted vessels landing over 1lb of longfin squid)	Below long term (1996-2023) average	Number of commercial vessels has been steadily decreasing since around 2000 consistent with decreasing fleet diversity and continued risk to fishery resilience [7]. Permit requalification in 2019 and a decrease in the post-closure trip limit for trimester 2 may cap participation, although these actions were designed to accommodate recent fishing trends and activity.	
Commercial revenue (millions, inflation adjusted to 2025 USD)	Near long term (1996-2023) average	Commercial revenue has decreased since 2022, driven by the overall decrease in landings by 23% [7].	
Western Gulf Stream Index (shift in the western part of the Gulf Stream North wall: mean position: >0 = more northerly, <0 = more southerly)	Above long term (1996-2023) average	Since the mid-1990s, north and westward shifts in the Gulf Stream have resulted in an increase in warm core rings and deep water, high salinity heat waves. The position of the Gulf Stream influences seasonal temperature and water mass mixing dynamics that affect longfin squid habitat suitability, temperature-dependent growth, and prey availability (https://noaa-edab.github.io/catalog/gsi.html).	
Bottom temperature in Mid-Atlantic Bight (MAB) and Southern New England (SNE)(°C)	Near long term (1996-2023) average (Fall and Spring)	Inshore temperature thresholds (around 14°C) initiate migration of squid from offshore overwintering habitats. Longfin squid seasonal distribution and growth rates are likely temperature dependent, avoiding water <8°C. Stronger and/or more persistent Mid-Atlantic Cold Pool conditions (not shown) may limit habitat availability (https://noaa-edab.github.io/catalog/cold_pool.html).	

Data Gaps/Uncertainty

- Bottom temperature data comes from GLORYS [8], a modeled re-analysis product that incorporates insitu data.
- The Western Gulf Stream Index indicator is a yearly value and may not be indicative of changes in oceanographic processes on a smaller time scale.
- Aging uncertainty creates uncertainty around life history processes, spawning location and timing, and natural mortality.
- While survey data in the 1970s and 80s indicate larval squid south of Cape Hatteras in the winter months that are transported north into the Mid-Atlantic Bight, there is a lack of definitive data to assess this hypothesis.
- Effects of cannibalism on the population are unknown at this time.

We welcome your observations! Please contact northeast.ecosystem.highlights@noaa.gov with any on-the-water insights or changes observed in the longfin inshore squid fishery and nefsc.esp.leads@noaa.gov with questions or comments on the information presented in this report.

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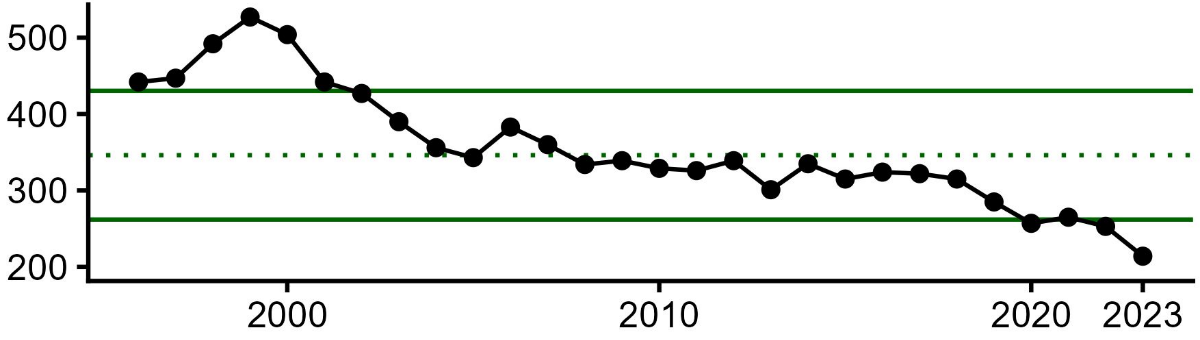
U.S. Department of Commerce | National Oceanic and Atmospheric Administration | National Marine Fisheries Service | Northeast Fisheries Science Center

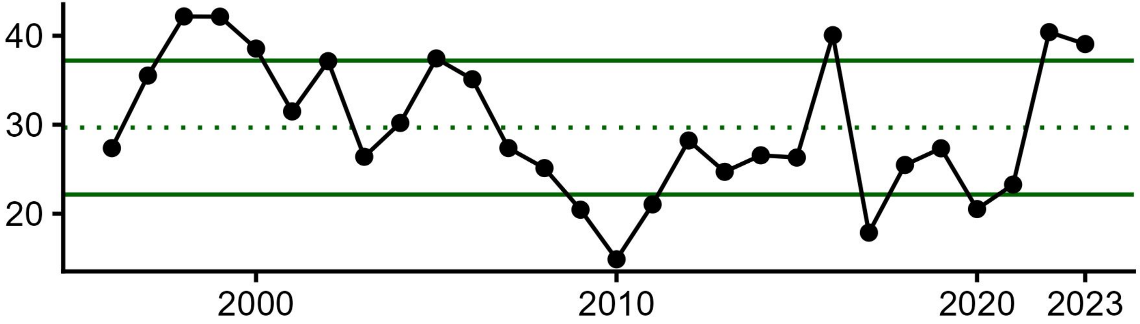
Number of Commercial Vessels

- Federally-permitted vessels landing >1lb of longfin
- Declines in vessels consistent with decreasing fleet diversity
- Permit requalification in 2019 and decrease in trip limits for trimester 2 after fishery closures
- Increase in landings but decline in vessels could indicate more targeted longfin trips



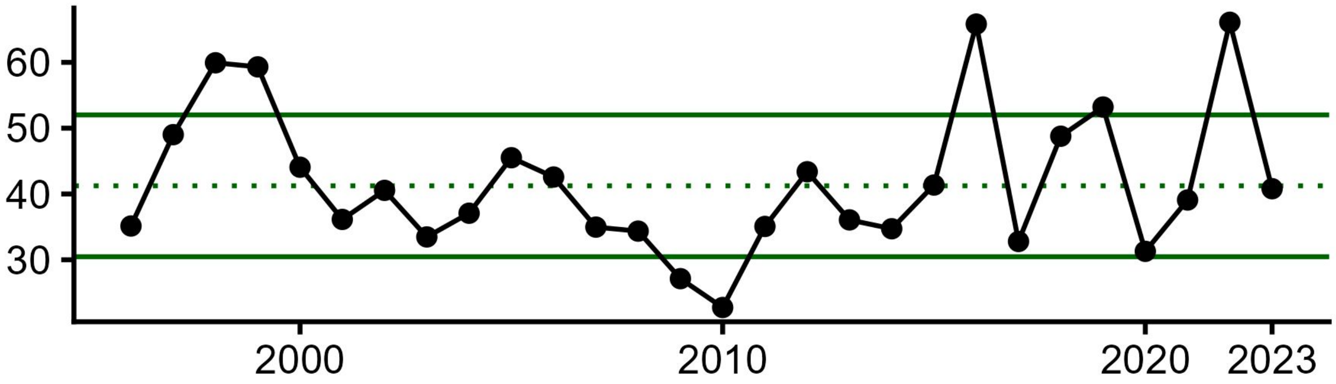
Commercial Landings (millions of lbs)

- Above long-term mean in 2023 despite decline in vessels
- Annual landings may be influenced by seasonal environmental dynamics in the summer and winter fisheries
- Increase in landings but decline in vessels could indicate more targeted longfin trips



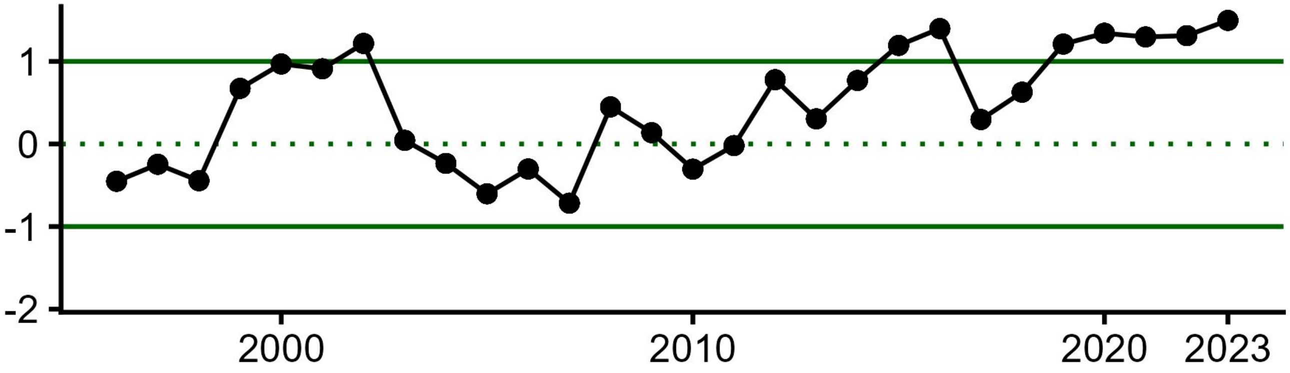
Commercial Revenue (millions USD$)

- Adjusted to inflation, 2025 USD
- Decline in revenue from 2022 to 2023, but near long-term mean
- May be driven by declines in price per pound



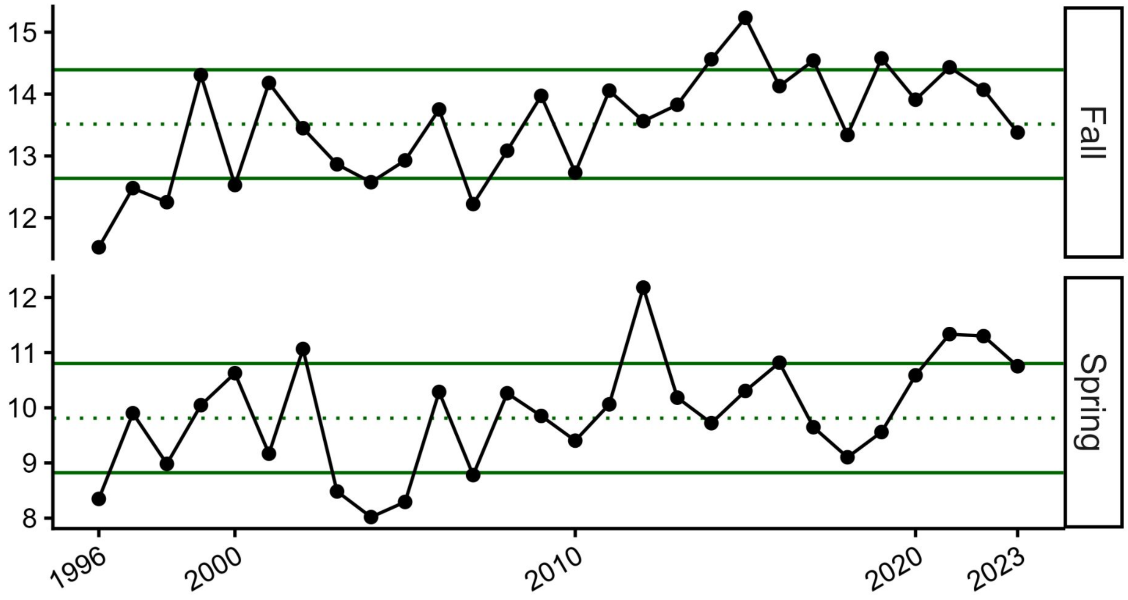
Western Gulf Stream Index

- Shift in the western part of the Gulf Stream North wall
- Mean position: >0 = more northerly, <0 = more southerly
- More northerly Gulf Stream position increases number of warm core rings and high salinity heat waves
- Gulf Stream influences seasonal temperature and water mass mixing that affect longfin habitat suitability



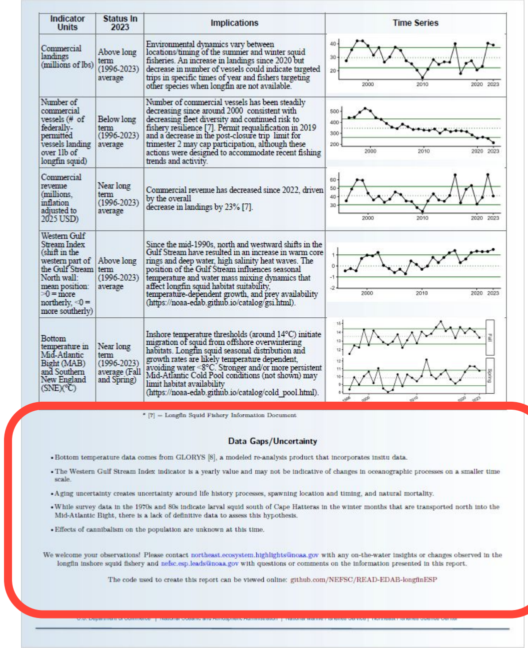
Bottom temperature (MAB and SNE)

- Fall and spring averages coinciding with timing of fishery
- Longfin avoid water <8C
- Temperature thresholds around 14C initiate inshore migration



Areas of Uncertainty

- Aging uncertainty creates uncertainty around life history processes, spawning location and timing, and natural mortality.
- Lack of definitive data to assess hypothesis surrounding south of Hatteras population inputs
- Effects of cannibalism on the population are unknown



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* [7] – Longfin Squid Fishery Information Document

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Thank you!

NEFSC Ecosystem Dynamics and Assessment Branch & ESP Team

NEFSC Social Sciences Branch

Longfin Squid RTA Working Group & Life History Subgroup

Squid Squad

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NOAA FISHERIES

Extra Figures